Q1

$f(x) = \frac{1}{2} x^T Q x - b^T x \qquad Q = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix} \qquad b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

(a) $Q^T = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix} = Q$

$\det |Q - \lambda I| = 0$

$\Rightarrow \begin{vmatrix} 4-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix} = 0$

$\Rightarrow (4-\lambda)(3-\lambda) - 1 = 0$

$12 - 7\lambda + \lambda^2 - 1 = 0$

$\lambda^2 - 7\lambda + 11 = 0$

$\lambda = \frac{7 \pm \sqrt{5}}{2} > 0$

$\Rightarrow Q$ is symmetric positive definite.

(b) $\nabla f(x) = Qx - b$

At minimizer: $\nabla f(x) = 0$

$\Rightarrow Qx' - b = 0$

$\Rightarrow Qx' = b$

$\Rightarrow x' = Q^{-1} b$

$\Rightarrow x' = \left(\frac{1}{11}\right) \begin{bmatrix} 3 & -1 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$\Rightarrow x' = \frac{1}{11}(1, 7)$

(c) level set: $f(x) = c$

$\frac{1}{2} x^T Q x - b^T x = c \qquad \begin{cases} Qx' = b \\ \Rightarrow b^T = x'^T Q^T = x'^T Q \end{cases}$

$\frac{1}{2} x^T Q x - x'^T Q x = c$

$\Rightarrow \left(\frac{x^T}{2} - x'^T\right) Q x = c$

$\Rightarrow$ level sets are ellipses.

(1) Eigenvalues $\lambda = \frac{7 \pm \sqrt{5}}{2}$

for $\lambda_1 = \frac{7 + \sqrt{5}}{2}$

$(Q - \lambda_1 I) x = 0$

$\Rightarrow \begin{pmatrix} 4 - \left(\frac{7+\sqrt{5}}{2}\right) & 1 \\ 1 & 3 - \left(\frac{7+\sqrt{5}}{2}\right) \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\Rightarrow \begin{pmatrix} 4-\lambda_1 & 1 \\ 1 & 3-\lambda_1 \end{pmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{\lambda_1} = \begin{bmatrix} \frac{-1+\sqrt{5}}{2} \\ 1 \end{bmatrix}$

for $\lambda_2 = \frac{7 - \sqrt{5}}{2}$

$(Q - \lambda_2 I) x = 0$

$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{\lambda_2} = \begin{bmatrix} \frac{1+\sqrt{5}}{2} \\ 1 \end{bmatrix}$

$\rightarrow \lambda_1$ is ~~larger~~ bigger eigenvalue

$\Rightarrow$ longer axis along $[x]_{\lambda_1}$

& shorter axis along $[x]_{\lambda_2}$

Q2 $x' = \begin{pmatrix} 1/11 \\ 7/11 \end{pmatrix}$

(a) $\nabla f(x) = Qx - b$

$x_{k+1} = x_k - \eta \nabla f(x_k)$

$= x_k - \eta (Qx_k - b)$

$x_{k+1} = [I - \eta Q] x_k + \eta b$

(b) error: $e_k = x_k - x'$ $\Rightarrow x_k = e_k + x'$

as $Qx' = b$ &

$x_{k+1} = [I - \eta Q] x_k + \eta b$

$\Rightarrow x_{k+1} - x' = [I - \eta Q] x_k + \eta b - x'$

$= [I - \eta Q](e_k + x') + \eta b - x'$

$= [I - \eta Q] x' + [I - \eta Q] e_k + \eta b - x'$

$= \cancel{x'} - \eta Q x' + [I - \eta Q] e_k + \eta b - \cancel{x'}$

$= \cancel{-\eta b} + [I - \eta Q] e_k \cancel{+ \eta b}$

$= [I - \eta Q] e_k$

$\Rightarrow e_{k+1} = [I - \eta Q] e_k$

(c) $e_k = (I - \eta Q^k) e_0$

Convergence $\Leftrightarrow (I - \eta Q)^k \to 0$

$\Leftrightarrow \rho(I - \eta Q) < 1$

↳ spectral radius of convergence

If $Qv = \lambda v \Rightarrow (I - \eta Q)v = (1 - \eta \lambda)v$

$\mu_i = 1 - \eta \lambda_i$

Hence, $\rho(I - \eta Q) = \max |1 - \eta \lambda_i|$

$|1 - \eta \lambda_i| < 1 \ \forall i$ (convergence condition)

$\Rightarrow -1 < 1 - \eta \lambda_i < 1$

$\eta \lambda_i < 2$ & $0 < \eta \lambda_i$

$\Rightarrow 0 < \eta < \dfrac{2}{\lambda_{max}}$

(d) Range of η.

$\lambda_{max} = \lambda_1 = \dfrac{7 + \sqrt{5}}{2}$

$\Rightarrow 0 < \eta < \dfrac{4}{7 + \sqrt{5}}$

$\Rightarrow 0 < \eta < 0.43$

(e) Asymptotic convergence factor:

$\rho(\eta) = \max \{|1 - \eta \lambda_{min}|, |1 - \eta \lambda_{max}|\}$

optimal η: $1 - \eta \lambda_{min} = -(1 - \eta \lambda_{max})$

$\Rightarrow 2 = \eta(\lambda_{min} + \lambda_{max})$

$\Rightarrow \eta_{opt} = 2/(\lambda_{min} + \lambda_{max})$

$= 2/(7) \approx 0.285$

$\Rightarrow \eta_{opt} \approx 0.285$

Q3 $e_{k+1} = (I - \eta Q) e_k$

(a)
(b) $K = \dfrac{d_{max}}{d_{min}}$

$f(\eta) = \max_i |1 - \eta d_i|$

$\eta_{opt} = \dfrac{2}{d_{max} + d_{min}}$

$1 - \eta_{opt} d_{max} = 1 - \left(\dfrac{2}{d_{max} + d_{min}}\right) d_{max}$

$= \dfrac{d_{min} - d_{max}}{d_{min} + d_{max}}$

$1 - \eta_{opt} d_{min} = 1 - \left(\dfrac{2}{d_{min} + d_{max}}\right) d_{min}$

$= \dfrac{d_{max} - d_{min}}{d_{min} + d_{max}}$

$\Rightarrow |1 - \eta_{opt} d_{max}| = |1 - \eta_{opt} d_{min}|$

$\Rightarrow f(\eta_{opt}) = \dfrac{d_{max} - d_{min}}{d_{max} + d_{min}}$

$= \dfrac{K - 1}{K + 1}$

(c) level set $f(x) = c$

$(x - x')^T Q (x - x') = c$

axes lengths - $\sqrt{\dfrac{c}{d_{min}}}$, $\sqrt{\dfrac{c}{d_{max}}}$

If $K >> 1$

ellipse becomes highly elongated.

$\hookrightarrow \max_i |1 - \eta d_i| = f(\eta)$

so, eigenvalue direction $\propto \dfrac{1}{\text{contraction}}$.

as $K \to \infty$, $f^* \to 1$

$\Rightarrow$ convergence slows down.

(d) let $y = P^T x$, wher P diagonalizes Q.

$f(x) = \frac{1}{2} y^T \Lambda y - (P^T b)^T y$

$y_{k+1} = y_k - \eta(\Lambda y_k - P^T b)$

$\hookrightarrow y_{k+1}^{(i)} = y_k^{(i)} - \eta(\Lambda y_k^{(i)} - \tilde{b}_i)$

$e_{k+1}^{(i)} = (1 - \eta d_i) e_k^{(i)}$

Q4 $K = 100 = \frac{\lambda_{max}}{\lambda_{min}}$

(a) let $\lambda_{min} = 1$, $\lambda_{max} = 100$

$Q = \begin{pmatrix} 100 & 0 \\ 0 & 1 \end{pmatrix}$

(b) $f(x) = \frac{1}{2} x^T Q x$

$x_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

error recursion:

$x_{k+1} = (I - \eta Q) x_k$

$\nabla f(x) = Qx$

~~$x_{k+1} =$~~ $\eta_{opt} = \frac{2}{\lambda_{max} + \lambda_{min}}$

$= \frac{2}{101} \approx 0.019$

$x_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$x_{k+1} = x_k - \alpha_k \nabla f(x_k)$

$= x_k - \alpha_k Q x_k$

$x_1 = x_0 - \alpha_k \begin{pmatrix} 100 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$= \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \alpha_k \begin{pmatrix} 100 \\ 1 \end{pmatrix}$

$= \frac{1 - \alpha_k 100}{1 - \alpha_k}$

(c) $\lambda_{max} \Rightarrow$ factor ≈ -0.98 $\left(f^* = \frac{99}{101} \right)$

$\lambda_{min} \Rightarrow$ factor ≈ 0.98

So, one coordinates flips sign each iteration.
then other decays slowly without flipping.
Thus alternates across narrow valley.
Level set is an elongated ellipse.
Hence, zig-zag trajectory.

(d) let $M = Q$

then, $M^{-1} Q x = M^{-1} b$

$Q_{pre} = M^{-1/2} Q M^{-1/2} = I$.

$K_{Q_{pre}} = \frac{\lambda_{max}}{\lambda_{min}} = \frac{1}{1} = 1$

$\Rightarrow$ perfect circles.

$\Rightarrow$ gradient descent converges in exactly one step.

Q5

(a) Lagrangian with two multipliers! d_1, d_2

$$L(x,y,z,d_1,d_2) = x^2+y^2+z^2+d_1(x+y+z-1) + d_2(x-y+2z)$$

(b) Full KKT system!

$$\frac{\partial L}{\partial x} = 0 \Rightarrow 2x + d_1 + d_2 = 0$$

$$\frac{\partial L}{\partial y} = 0 \Rightarrow 2y + d_1 - d_2 = 0$$

$$\frac{\partial L}{\partial z} = 0 \Rightarrow 2z + d_1 + 2d_2 = 0$$

Primal feasability:

$$x+y+z=1$$
$$x-y+2z=0$$

So, full KKT is!

$$\begin{cases} 2x+d_1+d_2=0 \\ 2y+d_1-d_2=0 \\ 2z+d_1+2d_2=0 \\ x+y+z=1 \\ x-y+2z=0 \end{cases}$$

(c) $x = -\left(\frac{d_1+d_2}{2}\right), y = \frac{d_2-d_1}{2},$

$$z = -\left(\frac{d_1+2d_2}{2}\right)$$

$\rightarrow x+y+z=1 \Rightarrow \frac{-3d_1+2d_2}{2}=1$

$\Rightarrow 3d_1+2d_2=-2$ — (i)

$\rightarrow x-y+2z=0 \Rightarrow d_1+3d_2=0$ — (ii)

By (i) & (ii), $d_1 = \frac{-6}{7}, d_2 = \frac{2}{7}$

$$x = -\left(\frac{d_1+d_2}{2}\right) = \frac{2}{7}$$

$$y = \frac{d_2-d_1}{2} = \frac{4}{7}$$

$$z = -\left(\frac{d_1+2d_2}{2}\right) = \frac{1}{7}$$

$\therefore (x', y', z') = \left(\frac{2}{7}, \frac{4}{7}, \frac{1}{7}\right)$

(d) $f(x) = x^2+y^2+z^2$

is strictly convex.

KKT conditions satisfied.

$\Rightarrow$ Solution is unique global minimizer

$x+y+z = \frac{2}{7}+\frac{4}{7}+\frac{1}{7} = 1$
$x-y+2z = \frac{2}{7}-\frac{4}{7}+\frac{2}{7} = 0$
Both satisfied

(e) $f(x,y,z) = x^2+y^2+z^2 = \|x\|^2$

Subject to $x+y+z=1$, $x-y+2z=0$ } line in 3D.

So, we choose point on line closest to origin.

$\Rightarrow$ Orthogonal projection of origin onto the affine subspace.

Q6 $f(x) = \frac{1}{2}x^TQx - b^Tx$

Q is SPD.

(a) $\nabla f(x) = Qx - b$

$\nabla^2 f(x) = Q$

as Q is SPD

$\Rightarrow v^TQv \geq \lambda_{min} ||v||^2$

$$f(y) = f(x) + \nabla f(x)^T(y-x) + \frac{1}{2}(y-x)^TQ(y-x)$$

$$(y-x)^TQ(y-x) \geq \lambda_{min} ||y-x||^2$$

So, $f(y) \geq f(x) + \nabla f(x)^T(y-x) + \frac{\lambda_{min}}{2}||y-x||^2$

So, $f(x)$ is strongly convex

(b) Gradient descent:

$$x_{k+1} = x_k - \eta(Qx_k - b)$$

error recursion:

$$e_{k+1} = (I - \eta Q)e_k$$

$$\rho = \max_i |1 - \eta \lambda_i|$$

Condition for convergence

$$|1 - \eta\lambda_i| < 1 \ \forall i$$

$$\Rightarrow \quad 0 < \eta < \frac{2}{\lambda_{max}}$$

Hence, gradient descent converges linearly.

(c) $e_k = (I - \eta Q)^k e_0$

$\Rightarrow ||e_k|| \leq \rho^k ||e_0||$

where, $\rho = \max_i |1 - \eta\lambda_i|$

~~$\Rightarrow ||e_k|| \leq$~~

as $e_k = x_k - x'$

$e_0 = x_0 - x'$

$$\Rightarrow ||x_k - x'|| \leq \rho^k ||x_0 - x'||$$

(d) Gradient Descent:

$$x_{k+1} = x_k - \eta(Qx_k - b)$$

Convergence rate: $\rho = \frac{K-1}{K+1}$

$\rightarrow$ Linear depends on K

Newton's method: $x_{k+1} = x_k - [\nabla^2 f(x_k)]^{-1} \nabla f(x_k)$

for quadratic: $\nabla^2 f(x) = Q$

Thus: $x_{k+1} = x_k - Q^{-1}(Qx_k - b)$

$$= x_k - x_k + Q^{-1}b$$

$$= Q^{-1}b = x'$$

Converges in one step.

Q7 (a) Total samples = 6

Class 1 = 3

Class 0 = 3

$p = \frac{3}{6} = 1/2$

$\cancel{G_{root} = 2 \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{2}}$

$\cancel{= 0.5}$

$\cancel{G_{root} = 0.5}$

$G_{root} = 1 - \sum p_i^2$

$= 1 - \left(\frac{1}{4} + \frac{1}{4}\right)$

$= \frac{1}{2} = 0.5$

(b) Splits on X_1:

$X_1 \leq 1.5$: $G = \frac{1}{6}(0) + \frac{5}{6}(0.48)$

$= 0.4$

$X_1 \leq 2.5$: $G = \frac{2}{6}(0) + \frac{4}{6}(0.375)$

$= 0.25$

$X_1 \leq 3.5$: $G = \frac{3}{6}(0.444) + \frac{3}{6}(0.444)$

$= 0.444$

$X_1 \leq 4.5$: $G = 0.5$

$X_1 \leq 5.5$: $G = 0.4$

(c) Splits on X_2:

$X_2 \leq 1.5$: $G = 0.4$

$X_2 \leq 2.5$: $G = 0$

$X_2 \leq 3.5$: $p = \frac{1}{4}$

$G = 2\left(\frac{1}{4}\right)\left(\frac{3}{4}\right) = \frac{3}{8} = 0.375$

Weighted: $\frac{4}{6}(0.375) = 0.25$

$X_2 \leq 4.5$: $G = 0.48$

weighted = 0.40

(d) Best split:

X_2 at 2.5 gives impurity 0.

So, best split $X_2 \leq 2.5$

Impurity reduction = $0.5 - 0 = 0.5$

(e) Root split: $X_2 \leq 2.5$

Left node: All $Y = 0 \rightarrow$ leaf $\rightarrow$ class 0

Right node: All $Y = 1 \rightarrow$ right $\rightarrow$ class 1

$X_2 \leq 2.5$

Yes   No

(f) Classify (4,2): $X_1 = 4, X_2 = 2$

Split: $X_2 \leq 2.5$

As $2 \leq 2.5$

$\Rightarrow$ (4,2) goes to left $\rightarrow$ class 0

Q11

(a)

	Tree impurity	Exponential loss	Logistic loss
Objective	Local greedy	Global	Global
Convex	No	Yes	Yes
Probablistic	No	No	Yes

(b) AdaBoost update: $\alpha_t = \frac{1}{2} \ln\left(\frac{1-\epsilon_t}{\epsilon_t}\right)$ & weight update: $w_i^{(t+1)} \propto w_i^{(t)} e^{-\alpha_t y_i h_t(x_i)}$

let $F_T(x) = \sum \alpha_t h_t(x) \Rightarrow w_i^{(T)} \propto e^{-y_i F_T(x_i)}$

$L(F) = \sum_i e^{-y_i F(x_i)}$ ∴ AdaBoost performs coordinate descent on $L(F)$.

Each step minimizes this loss ⇒ AdaBoost minimizes exponential loss.

(c) Binary Model: $p_i = \sigma(w^T x_i)$

log-likelihood: $l(w) = \sum_i [y_i \log p_i + (1-y_i)\log(1-p_i)]$

Negative log-likelihood: $L(w) = \sum_i \log(1 + e^{-y_i w^T x_i})$

Thus, logistic regression minimizes ~~logistic~~ (log) loss.

(d) $L(F) = \sum_i l(y_i, F(x_i))$

At step t: $r_i^{(t)} = -\frac{\partial l(y_i, F(x_i))}{\partial F(x_i)}\Big|_{F = F_{t-1}}$ & Fit weak learner $h_t(x) \to r_i^{(t)}$

update: $F_t(x) = F_{t-1}(x) + \eta h_t(x)$ ⇒ Gradient descent in function space.

Thus, gradient boosting performs functional gradient descent.

Q10

(a) $l[w,b] = \sum_{i=1}^{4}[y_i \log(\sigma(z_i)) + (1-y_i)\log(1-\sigma(z_i))]$

$$\log(\sigma(z)) = \log([1+e^{-z}]^{-1}) = -\log(1+e^{-z}) = z - \log(1+e^{z})$$

$$\log(1-\sigma(z)) = \log\left(\frac{e^{-z}}{(1+e^{-z})}\right) = -z - \log(1+e^{-z}) = -z+z-\log(1+e^{z})$$

$$= -\log(1+e^{z})$$

$$l[w,b] = \sum_{i=1}^{4}\left[y_i(z_i - \log(1+e^{z_i})) + (1-y_i)(-\log(1+e^{z_i}))\right]$$

$$= \sum_{i=1}^{4}\left[y_i z_i - \log(1+e^{z_i})\right]$$

(b) $\frac{d\sigma(z)}{dz} = \frac{d}{dz}\left[\frac{1}{1+e^{-z}}\right] = \frac{e^{-z}}{(1+e^{-z})^2} = \left(\frac{1}{1+e^{-z}}\right)\left(\frac{e^{-z}}{1+e^{-z}}\right) = \sigma(z)(1-\sigma(z))$

So, $\frac{\partial l[w,b]}{\partial w} = \sum_{i=1}^{4} x_i[y_i - \sigma(z_i)]$

& $\frac{\partial l[w,b]}{\partial b} = \sum_{i=1}^{4}[y_i - \sigma(z_i)]$

(c) Start at (0,0): $w=0, b=0 \Rightarrow \sigma(0) = 0.5$

x	y	$y-\sigma$
0	0	-0.5
1	0	-0.5
2	1	0.5
3	1	0.5

$$\left.\frac{\partial l}{\partial w}\right|_{(0,0)} = 0(-0.5) + 1(-0.5) + 2(0.5) + 3(0.5)$$

$$= -0.5 + 1 + 1.5 = 2$$

$$\left.\frac{\partial l}{\partial b}\right|_{(0,0)} = -0.5 - 0.5 + 0.5 + 0.5 = 0$$

So, $\nabla l = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

$$w^{(1)} = w^{(0)} + \eta \nabla l = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$w^{(1)} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

(d) $H = -\sum_i \sigma(z_i)(1-\sigma(z_i)) \begin{bmatrix} x_i^2 & x_i \\ x_i & 1 \end{bmatrix}$

At $(w,b)=(0,0)$ $\quad \sigma(0)=0.5,\ 1-\sigma(0)=0.5$

$\Rightarrow \sigma(0)(1-\sigma(0)) = 0.25$

So, $H = -0.25 \sum \begin{bmatrix} x_i^2 & x_i \\ x_i & 1 \end{bmatrix}$

$= -0.25 \begin{bmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & \sum 1 \end{bmatrix}$

$\sum x_i^2 = 0+1+4+9 = 14,\ \sum x_i = 6,\ \sum 1 = 4$

$H = -0.25 \begin{bmatrix} 14 & 6 \\ 6 & 4 \end{bmatrix} = \begin{bmatrix} -3.5 & -1.5 \\ -1.5 & -1 \end{bmatrix}$

(e) $\theta^{(1)} = \theta^{(0)} - H^{-1}\nabla l \quad , \theta=(w,b)$

At $(0,0)$, $\quad H = \begin{bmatrix} -3.5 & -1.5 \\ -1.5 & -1 \end{bmatrix} \quad , \nabla l = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \quad , \theta^{(0)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\theta^{(1)} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} -0.8 & 1.2 \\ 1.2 & 2.8 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} -1.6 \\ 2.4 \end{bmatrix}$

$\theta^{(1)} = \begin{bmatrix} +1.6 \\ -2.4 \end{bmatrix}$

(f) $z = 1.6(1.5) - 2.4 \qquad \left| \begin{array}{l} w=1.6 \\ x=1.5,\ b=-2.4 \end{array} \right.$

$= 2.4 - 2.4 = 0$

$\sigma(0) = 0.5$

~~Pdy~~ $p(y=1 \mid x=1.5) = \sigma(0) = 0.5$

$wx+b=0 \Rightarrow x = \dfrac{-b}{w} = \dfrac{2.4}{1.6} = 1.5$

$\Rightarrow x=1.5$ lies on boundary.

Classification: $y=1$ at $x=1.5$

Q12

(a) $\frac{d}{dz}\log(\sigma(z)) = 1-\sigma(z)$

$\frac{d}{dz}\log(1-\sigma(z)) = -\sigma(z)$

$$\frac{\partial l}{\partial w} = \sum_{i=1}^{4} w_i x_i (y_i - \sigma(z_i))$$

$$\frac{\partial l}{\partial b} = \sum_{i=1}^{4} w_i (y_i - \sigma(z_i))$$

(b) $\frac{d\sigma(z)}{dz} = \sigma(z)(1-\sigma(z))$

$$H = -\sum_{i=1}^{4} w_i \sigma(z_i)(1-\sigma(z_i)) \begin{bmatrix} x_i^2 & x_i \\ x_i & 1 \end{bmatrix} = \begin{bmatrix} -12.5 & -5 \\ -5 & -2.5 \end{bmatrix}$$

(c) $(w,b) = (0,0)$

$z_i = 0 \Rightarrow \sigma(0) = 0.5$

x	y	w	$y-\sigma$	$(y-\sigma)$ weighted
0	0	1	-0.5	-0.5
1	0	2	-0.5	-1
2	1	3	0.5	1.5
3	1	4	0.5	2

$$\left.\frac{\partial l}{\partial w}\right|_{(0,0)} = 0(-0.5)+1(-1)+2(1.5)+3(2) = -1+3+6 = 8$$

$$\left.\frac{\partial l}{\partial b}\right|_{(0,0)} = -0.5-1+1.5+2 = 2$$

$\nabla l = (8,2)$

$$w^{(1)} = w^{(0)} + \eta \nabla l = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \eta \begin{bmatrix} 8 \\ 2 \end{bmatrix}$$

$\eta = 0.1$

$$w^{(1)} = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}$$

(d) Hight weight $\Rightarrow$ increased gradient influence

$\rightarrow$ increased curvature contribution

$\rightarrow$ Pulls boundary towards heavily weighted samples

Points with $(y=1)$ have higher weight.

So, decision boundary shifts left to classify positive class with more buffer.

(e) weighted likelihood: $l = \sum w_i \log p(y_i/x_i)$

$\sim$ Importance weighted maximum likelihood.

Q13 (a) $\bar{y} = \frac{\sum y_i}{6} = \frac{2+3+2+6+7+8}{6} = \frac{28}{6} = 4.6667$

$SSE = \sum_{i=1}^{6} (y_i - \bar{y})^2 = 35.333$

(b) Splits (midpoints): 1.5, 2.5, 3.5, 4.5, 5.5

Split: 1.5

$\rightarrow$ left: $\{2\} \rightarrow SSE = 0$

$\rightarrow$ Right: $\{3,2,6,7,8\}$ Mean $= 5.2$

$SSE \approx 26.8$

Total $= 26.8$

Split 2.5: $\rightarrow$ left: $\{2,3\}$

mean $= 2.5$

$SSE = (2-2.5)^2 + (3-2.5)^2$

$= 0.25 + 0.25 = 0.5$

$\rightarrow$ Right: $\{2,6,7,8\}$

Mean $= 5.75$

$SSE = 22.75$

Total ≈ 23.75

Split: 3.5 $\rightarrow$ left: $\{2,3,2\}$

Mean $= 2.33$

$SSE \approx 0.67$

$\rightarrow$ Right: $\{6,7,8\}$

mean $= 7$, $SSE \approx 2$

Total $= 0.67 + 2 = 2.67$

Split 4.5 $\rightarrow$ left: $\{2,3,2,6\}$

Mean $= 3.25$

$SSE \approx 12.75$

$\rightarrow$ Right: $\{7,8\}$

Mean $= 7.5$

$SSE = 0.5$

Total ≈ 13.25

Split 5.5:

Left: {2, 3, 2, 6, 7}

Mean = 4

SSE ≈ 19

Right: {8} → 0

Total ≈ 19

ⓒ Optimal First split:

Minimal SSE = 2.67

Best split at X = 3.5

~~Variance~~

ⓓ Split at 2.5: left {2, 3}

mean = 2.5

SSE = 0.5

Right {2}

mean = 2

SSE = 0

Total = 0.5 < 0.67

for right node:

Split at 5.5: left {6, 7}

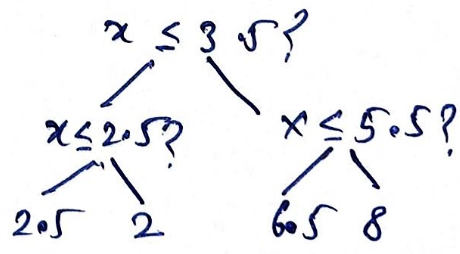
mean = 6.5

SSE = 0.5

Right {8}

SSE = 0

Total = 0.5 < 2



(e) at $X = 4.5$

$4.5 > 3.5 \rightarrow$ right

$4.5 < 5.5 \rightarrow$ left

So Prediction $(X = 4.5) = 6.5$

(f) Minimize!

$SSE = \sum (y - \bar{y})^2$ } minimize variance

for regression tree

for classification!

minimize $G = 2p(1-p)$ } minimize class impurity

Q8 (a) Possible thresholds 1.5, 2.5, 3.5

Threshold 1.5!

X	Y	h(X)
1	1	-1
2	1	1
3	-1	1
4	-1	1

$$h(x) = \begin{cases} -1 & x \le 1.5 \\ 1 & \text{otherwise} \end{cases}$$

weighted error $= \frac{3}{4}$! flip sign

X	Y	-h(X)
1	1	1
2	1	-1
3	-1	-1
4	-1	-1

weighted error $= \frac{1}{4}$

threshold 2.5!

$$h(x) = \begin{cases} -1 & x \le 2.5 \\ 1 & \text{otherwise} \end{cases}$$

X	Y	h(X)
1	1	-1
2	1	-1
3	-1	1
4	-1	1

weighted error = 1

flip sign!

X	Y	-h(X)
1	1	1
2	1	1
3	-1	-1
4	-1	-1

weighted error = 0

threshold 3.5 :

$$h(x) = \begin{cases} -1 & x \le 3.5 \\ 1 & \text{otherwise} \end{cases}$$

x	y	h(x)
1	1	-1
2	1	-1
3	-1	-1
4	-1	1

weighted error $= \frac{3}{4}$

after flip :

x	y	-h(x)
1	1	1
2	1	1
3	-1	1
4	-1	-1

weighted error $= \frac{1}{4}$

$$\boxed{\text{Best weak learner : } t = 2.5,\ \epsilon_1 = 0}$$

(b) $\alpha_1 = \frac{1}{2} \ln\left(\frac{1-\epsilon_1}{\epsilon_1}\right)$ $\quad \epsilon_1 = 0$

$\Rightarrow \alpha_1 \to \infty$

(c) $w_i^{(2)} = w_i^{(1)} e^{-\alpha_1 y_i h_1(x_i)}$ $\quad$ as $\alpha_1 \to \infty$

$\Rightarrow w_i^{(2)} = 0$

(d) $Z_1 = \sum_i w_i^{(2)} = 0$ $\quad$ Boosting stops

(e) $\epsilon_1 = 0,\ Z_1 = 0 \Rightarrow$ Algorithm terminates

(f) $F(x) = \alpha_1 h_1(x)$ $\quad$ as $\alpha_1 > 0 \Rightarrow H(x) = \text{sign}(F(x)) = h_1(x)$

$$\Rightarrow H(x) = \begin{cases} 1 & x \le 2.5 \\ -1 & x > 2.5 \end{cases}$$

Q9 (a) $\bar{y} = \frac{\sum y_i}{4} = \frac{19}{4} = 4.75$

$F_0(x) = 4.75$

(b) $r_i^{(1)} = y_i - 4.75$

x	y	$y_i - 4.75$
1	2	-2.75
2	3	-1.75
3	6	1.25
4	8	3.25

(c) Possible splits: 1.5, 2.5, 3.5

Split at 1.5: left $\{-2.75\} \rightarrow SSE = 0$

right $\{-1.75, 1.25, 3.25\}$

$\text{mean} = \frac{-1.75 + 1.25 + 3.25}{3} = 0.9167$

$SSE \approx 12.67$

$SSE\ Total = 12.67$

Split at 2.5: left $\{-2.75, -1.75\}$

$\text{mean} = -2.25$

$SSE = (-0.5)^2 + (0.5)^2 = 0.5$

Right $\{1.25, 3.25\}$

$\text{mean} = 2.25$

$SSE = (-1)^2 + (1)^2 = 2$

$SSE_{Total} = 2.5$

Split at 3.5: left $\{-2.75, -1.75, 1.25\}$

$\text{mean} = -1.08$

$SSE = (-2.75 + 1.08)^2 + (-1.75 + 1.08)^2 + (1.25 - 1.08)^2$

$= 8.6666$

Best split: at $X = 2.5$

$$h_1(x) = \begin{cases} -2.25 & x \leq 2.5 \\ 2.25 & x > 2.5 \end{cases}$$

(d) $F_1(x) = F_0(x) + \eta h_1(x)$

$\eta = 0.5$, $h_1(x) = \begin{cases} -2.25 & x \le 2.5 \\ 2.25 & x > 2.5 \end{cases}$, $F_0(x) = 4.75$

$$F_1(x) = \begin{cases} 4.75 - (0.5)(-2.25) & x \le 2.5 \\ 4.75 + (0.5)(2.25) & x > 2.5 \end{cases}$$

$$F_1(x) = \begin{cases} 3.625 & x \le 2.5 \\ 5.875 & x > 2.5 \end{cases}$$

(e) $r_i^{(2)} = y_i - F_1(x_i)$

x_i	$r_i^{(2)}$
1	-1.625
2	-0.625
3	0.125
4	2.125

split at $x = 2.5$ left $= \{-1.625, -0.625\}$
mean $= -1.125$
SSE $= 0.5$
right $= \{0.125, 2.125\}$
mean $= 1.125$
SSE $= 2$
Total $= 2.5$

$$h_2(x) = \begin{cases} -1.125 & x \le 2.5 \\ 1.125 & x > 2.5 \end{cases}$$

$$F_2(x) = F_1(x) + \eta_1 h_2(x)$$

$$= \begin{cases} 3.625 - 0.5625 \\ 5.875 + 0.5625 \end{cases}$$

$$F_2(x) = \begin{cases} 3.0625 & x \le 2.5 \\ 6.4375 & x > 2.5 \end{cases}$$

(f)

x	y	$F_2(x)$
1	2	3.0625
2	3	3.0625
3	6	6.4375
4	8	6.4375